



**IDC216 : Statistics for Scientists (2025-26)**

**Exercise Set 2**

1. In a lab, the probability that a sensitive detector works on a given day is 0.9. If the detector is tested for 5 days independently, find:
  - (a) The probability it works on all days.
  - (b) The probability it fails on exactly one day.
  - (c) The expected number of days when it works.
2. In a physics experiment, the error in measuring the length of a metal rod is normally distributed with mean 0 mm and standard deviation 2 mm.
  - (a) What is the probability that a single measurement error lies between  $-1$  mm and  $+2$  mm?
  - (b) What is the probability that the error exceeds  $+4$  mm?
  - (c) If 25 independent measurements are taken, what is the probability that the *average error* lies within  $\pm 1$  mm?
3. In a survey at a science institute, 70% of students reported using the library regularly. Among these, 60% scored above 80 in their course. Among those who do not use the library regularly, only 30% scored above 80.
  - (a) What is the probability that a randomly chosen student scored above 80?
  - (b) If a student scored above 80, what is the probability that they use the library regularly?
4. A pair of dice is rolled once. Define the following events:
  - $A$ : the sum of the number appearing on the two dice is an even number.
  - $B$ : the product of the number appearing on the two dice is an even number.
  - (a) Write down the event sets corresponding to  $A$  and  $B$ .
  - (b) Compute  $P(A)$  and  $P(B)$ .
  - (c) Determine if  $A$  and  $B$  are independent events.
  - (d) Now define another event  $C$ : the sum is a multiple of 3. Determine if  $A$  and  $C$  are independent.
5. The weights of lab rats are normally distributed with mean 250 g and standard deviation 20 g.
  - (a) What proportion of rats weigh more than 270 g?

- (b) Find the weight that marks the 90th percentile.
- (c) If a sample of 25 rats is selected, find the probability that their mean weight exceeds 260 g.
6. The diameter of a cricket ball follows a distribution with mean 70 mm and standard deviation 5 mm. The distribution is not known. A cricket coach inspects a random sample of  $n = 64$  balls. Using the central limit theorem, estimate the probability that the sample mean diameter lies between 68 mm and 71 mm.
7. The following table shows study hours ( $X$ ) and exam scores ( $Y$ ) for 6 students:

| $X$ | $Y$ |
|-----|-----|
| 2   | 50  |
| 3   | 45  |
| 4   | 55  |
| 5   | 70  |
| 6   | 80  |
| 7   | 75  |

- (a) Compute the coefficient of correlation.
- (b) Instead you rank the data and compute Spearman's rank correlation.

On this basis do you conclude that studying more improves exam scores?

8. In a class of 255 students, the following table shows enrollment in IDC216 vs. Math background in Class XII:

|                | Enrolled in IDC216 | Not Enrolled | Total |
|----------------|--------------------|--------------|-------|
| Math in XII    | 130                | 40           | 170   |
| No Math in XII | 32                 | 53           | 85    |
| Total          | 162                | 93           | 255   |

- (a) Compute the expected frequencies under independence.
- (b) Calculate the chi-square statistic.
- (c) At  $\alpha = 0.05$ , test whether enrollment depends on math background.
9. A survey of 180 students at IISER Mohali classified students according to their participation in sports in Class XII and their choice of major (Biology, Chemistry, Mathematics). The observed frequencies are:

|                  | Biology | Chemistry | Mathematics | Total |
|------------------|---------|-----------|-------------|-------|
| Sports in XII    | 20      | 50        | 30          | 100   |
| No sports in XII | 25      | 15        | 40          | 80    |
| Total            | 45      | 65        | 70          | 180   |

Note: *EES students were on a field trip and Physics student had a lab exam when this survey was conducted.*

- (a) Compute the expected frequencies under independence.

- (b) Calculate the chi-square test statistic.
- (c) With  $\alpha = 0.05$ , test whether elective choice depends on the sports activity taken in class XII.
10. A survey of 150 students asks about *gender* (Male/Female) and preferred mode of study (Textbooks, Online videos, Group discussions). The results are:

|        | Textbooks | Online Videos | Group Discussions | Total |
|--------|-----------|---------------|-------------------|-------|
| Male   | 25        | 30            | 20                | 75    |
| Female | 35        | 15            | 25                | 75    |
| Total  | 60        | 45            | 45                | 150   |

Compute the chi-square statistic and test independence between gender and study mode.